

Sunghwan Yoo

Computer Vision & AI Research and Development
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Education

- Gachon University — M.S. student, School of Computing**

Sep. 2026 – Present
- CVIP Lab, Seongnam, South Korea.
- Gachon University — B.S., School of Computing**

Mar. 2021 – Aug. 2026
- GPA: 4.47/4.5 overall; 4.5/4.5 major. Graduated first in major with early graduation.

Technical Skills

- Tools:** Python, PyTorch, OpenCV, YOLO, ONNX.
- Areas:** Infrared-visible image fusion, image deblurring, object detection, event-based depth reconstruction.

Experience

- CVIP Lab, Gachon University**

Dec. 2023 – Present
- M.S. student since Sep. 2026; previously Undergraduate Research Assistant.
- Research image fusion and deblurring; contributed to three KCC conference papers and two journal manuscripts currently under review.
- Freelance Computer Vision Engineer — Freemoa**

Feb. – Mar. 2025
- Developed a YOLO-based model to locate cattle tails and classify raised/lowered states in barn CCTV footage.
 - Redesigned labeling criteria, improved training data, analyzed false detections from barn structures, applied OpenCV augmentation, tuned models, and delivered an ONNX model.

Research and Industry Projects

- Private 5G Military Communications — LIG Collaboration**

Mar. 2025 – Present
- Researched defense applications and technology trends, planned tactical scenarios, and prepared technical reports using M&S results supplied by a collaborating lab.
- DVS-based 3D Depth Reconstruction — NRF Mid-Career Research**

Sep. 2025 – Present
- Research AI algorithms for high-quality 3D depth reconstruction from dynamic vision sensors (DVS).
- Advanced GPU Utilization Support Program — MSIT/NIPA**

Apr. – Jul. 2026
- Prepared technical proposals and research reports, analyzed training compute requirements, and planned GPU utilization. Contributed to the application's successful selection.

Selected Publications

- Infrared-Visible Image Fusion via Heat Conduction-based Frequency Fusion.** First author, KCC 2026.
- HeatFuse: Heat-Conduction-Based Spectral Decomposition and Cross-Band Modulation for Infrared-Visible Image Fusion.** Under review, *IEEE Trans.*
- Cross-stage Patch Interaction for Coarse-to-Fine Image Deblurring.** Under review, *MDPI*.

Honors

- KCC 2026 Undergraduate Paper Competition — Encouragement Award (Aug. 2026).
- National Science and Engineering Excellence Scholarship, Korea Student Aid Foundation (Oct. 2025).